

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ nano pipeline.sh
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ ls
SRR27182345_1.fastq.gz  SRR27182345_2.fastq.gz  chr7.fa.gz  pipeline.sh
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ gunzip SRR27182345_1.fastq.gz  SRR27182345_2.fastq.gz  chr7.fa.gz
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ ls
SRR27182345_1.fastq  SRR27182345_2.fastq  chr7.fa  pipeline.sh
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ fastqc SRR27182345_1.fastq  SRR27182345_2.fastq
null
Started analysis of SRR27182345_1.fastq
null
Approx 5% complete for SRR27182345_1.fastq
Approx 10% complete for SRR27182345_1.fastq
Approx 15% complete for SRR27182345_1.fastq
Approx 20% complete for SRR27182345_1.fastq
Approx 25% complete for SRR27182345_1.fastq
Approx 30% complete for SRR27182345_1.fastq
Approx 35% complete for SRR27182345_1.fastq
Approx 40% complete for SRR27182345_1.fastq
Approx 45% complete for SRR27182345_1.fastq
Approx 50% complete for SRR27182345_1.fastq
Approx 55% complete for SRR27182345_1.fastq
Approx 60% complete for SRR27182345_1.fastq
Approx 65% complete for SRR27182345_1.fastq
Approx 70% complete for SRR27182345_1.fastq
Approx 75% complete for SRR27182345_1.fastq
Approx 80% complete for SRR27182345_1.fastq
Approx 85% complete for SRR27182345_1.fastq
Approx 90% complete for SRR27182345_1.fastq
Approx 95% complete for SRR27182345_1.fastq
Analysis complete for SRR27182345_1.fastq
Started analysis of SRR27182345_2.fastq
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ fastp -i SRR27182345_1.fastq -o trim_SRR27182345_1.fastq -I SRR27182345_2.fastq -O trim_SRR27182345_2.fastq -5 -3 -q 30 --adapter_fasta adapter.fasta.txt
Read1 before filtering:
total reads: 70108
total bases: 10564958
Q20 bases: 10089867(95.5031%)
Q30 bases: 9943095(94.1139%)

Read2 before filtering:
total reads: 70108
total bases: 10538945
Q20 bases: 9850677(93.4693%)
Q30 bases: 9647177(91.5384%)

Read1 after filtering:
total reads: 66824
total bases: 9433500
Q20 bases: 9141966(96.9096%)
Q30 bases: 9029257(95.7148%)

Read2 after filtering:
total reads: 66824
total bases: 9119266
Q20 bases: 8829785(96.8256%)
Q30 bases: 8691210(95.306%)

Filtering result:
reads passed filter: 133648
reads failed due to low quality: 3986
reads failed due to too many N: 78
reads failed due to too short: 2504
reads with adapter trimmed: 22866
bases trimmed due to adapters: 1223491

Duplication rate: 71.9604%

Insert size peak (evaluated by paired-end reads): 181

JSON report: fastp.json
HTML report: fastp.html
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ fastqc trim_SRR27182345_1.fastq trim_SRR27182345_2.fastq
null
Started analysis of trim_SRR27182345_1.fastq
null
Approx 5% complete for trim_SRR27182345_1.fastq
Approx 10% complete for trim_SRR27182345_1.fastq
Approx 15% complete for trim_SRR27182345_1.fastq
Approx 20% complete for trim_SRR27182345_1.fastq
Approx 25% complete for trim_SRR27182345_1.fastq
Approx 30% complete for trim_SRR27182345_1.fastq
Approx 35% complete for trim_SRR27182345_1.fastq
Approx 40% complete for trim_SRR27182345_1.fastq
Approx 45% complete for trim_SRR27182345_1.fastq
Approx 50% complete for trim_SRR27182345_1.fastq
Approx 55% complete for trim_SRR27182345_1.fastq
Approx 60% complete for trim_SRR27182345_1.fastq
Approx 65% complete for trim_SRR27182345_1.fastq
Approx 70% complete for trim_SRR27182345_1.fastq
Approx 75% complete for trim_SRR27182345_1.fastq
Approx 80% complete for trim_SRR27182345_1.fastq
Approx 85% complete for trim_SRR27182345_1.fastq
Approx 90% complete for trim_SRR27182345_1.fastq
Approx 95% complete for trim_SRR27182345_1.fastq
Analysis complete for trim_SRR27182345_1.fastq
Started analysis of trim_SRR27182345_2.fastq
```

```
krishn@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ bwa index -a bwtsw chr7.fa
[bwa_index] Pack FASTA... 1.16 sec
[bwa_index] Construct BWT for the packed sequence...
[BWTIncCreate] textLength=318691946, availableWord=34423912
[BWTIncConstructFromPacked] 10 iterations done. 56784042 characters processed.
[BWTIncConstructFromPacked] 20 iterations done. 104904522 characters processed.
[BWTIncConstructFromPacked] 30 iterations done. 147670090 characters processed.
[BWTIncConstructFromPacked] 40 iterations done. 185676202 characters processed.
[BWTIncConstructFromPacked] 50 iterations done. 219452122 characters processed.
[BWTIncConstructFromPacked] 60 iterations done. 249468234 characters processed.
[BWTIncConstructFromPacked] 70 iterations done. 276142554 characters processed.
[BWTIncConstructFromPacked] 80 iterations done. 299846746 characters processed.
[bwt_gen] Finished constructing BWT in 89 iterations.
[bwa_index] 91.53 seconds elapse.
[bwa_index] Update BWT... 0.70 sec
[bwa_index] Pack forward-only FASTA... 0.84 sec
[bwa_index] Construct SA from BWT and Occ... 50.06 sec
[main] Version: 0.7.17-r1188
[main] CMD: bwa index -a bwtsw chr7.fa
[main] Real time: 151.778 sec; CPU: 144.294 sec
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ bwa mem -t 9 chr7.fa trim_SRR27182345_1.fastq trim_SRR27182345_2.fastq > bwa_SRR27182345.sam
[M::bwa_idx_load_from_disk] read 0 ALT contigs
[M::process] read 133648 sequences (18552766 bp)...
[M::mem_pestat] # candidate unique pairs for (FF, FR, RF, RR): (0, 6040, 0, 0)
[M::mem_pestat] skip orientation FF as there are not enough pairs
[M::mem_pestat] analyzing insert size distribution for orientation FR...
[M::mem_pestat] (25, 50, 75) percentile: (250, 250, 250)
[M::mem_pestat] low and high boundaries for computing mean and std.dev: (250, 250)
[M::mem_pestat] mean and std.dev: (250.00, 0.00)
[M::mem_pestat] low and high boundaries for proper pairs: (250, 250)
[M::mem_pestat] skip orientation RF as there are not enough pairs
[M::mem_pestat] skip orientation RR as there are not enough pairs
[M::mem_process_seqs] Processed 133648 reads in 49.871 CPU sec, 12.534 real sec
[main] Version: 0.7.17-r1188
[main] CMD: bwa mem -t 9 chr7.fa trim_SRR27182345_1.fastq trim_SRR27182345_2.fastq
[main] Real time: 20.615 sec; CPU: 50.905 sec
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ samtools sort bwa_SRR27182345.sam > sorted_SRR27182345.sam
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ ls
```

SRR27182345_1.fastq	SRR27182345_2_fastqc.html	chr7.fa	chr7.fa.pac	pipeline.sh	trim_SRR27182345_1_fastqc.zip
SRR27182345_1_fastqc.html	SRR27182345_2_fastqc.zip	chr7.fa.amb	chr7.fa.sa	sorted_SRR27182345.sam	trim_SRR27182345_2.fastq
SRR27182345_1_fastqc.zip	adapter.fasta.txt	chr7.fa.ann	fastp.html	trim_SRR27182345_1.fastq	trim_SRR27182345_2_fastqc.html
SRR27182345_2.fastq	bwa_SRR27182345.sam	chr7.fa.bwt	fastp.json	trim_SRR27182345_1_fastqc.html	trim_SRR27182345_2_fastqc.zip


```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ samtools rmdup -s SRR27182345.sam rmdup_SRR27182345.sam
[bam_rmdupse_core] 26978 / 29782 = 0.9058 in library ' '
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ ls
SRR27182345.sam          SRR27182345_2_fastqc.html  chr7.fa.amb  fastp.html          trim_SRR27182345_1.fastq      trim_SRR27182345_2_fastqc.zip
SRR27182345_1.fastq      SRR27182345_2_fastqc.zip   chr7.fa.ann  fastp.json          trim_SRR27182345_1_fastqc.html
SRR27182345_1_fastqc.html adapter.fasta.txt          chr7.fa.bwt  pipeline.sh         trim_SRR27182345_1_fastqc.zip
SRR27182345_1_fastqc.zip bwa_SRR27182345.sam        chr7.fa.pac  rmdup_SRR27182345.sam trim_SRR27182345_2.fastq
SRR27182345_2.fastq      chr7.fa                    chr7.fa.sa   sorted_SRR27182345.sam trim_SRR27182345_2_fastqc.html
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$
```



```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ picard-tools CreateSequenceDictionary R=chr7.fa O=chr7.dict
```

```
JavOpt:
```

```
PicardProg: CreateSequenceDictionary
```

```
PicardOpts: R=chr7.fa O=chr7.dict
```

```
Warning: picard-tools is deprecated and should be replaced by PicardCommandLine
```

```
INFO      2026-07-29 10:55:39      CreateSequenceDictionary
```

```
***** NOTE: Picard's command line syntax is changing.
```

```
*****
```

```
***** For more information, please see:
```

```
*****
```

```
https://github.com/broadinstitute/picard/wiki/Command-Line-Syntax-Transition-For-Users-\(Pre-Transition\)
```

```
*****
```

```
***** The command line looks like this in the new syntax:
```

```
*****
```

```
*****      CreateSequenceDictionary -R chr7.fa -O chr7.dict
```

```
*****
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/Crispr$ picard-tools AddOrReplaceReadGroups I=rmDup_SRR27182345.sam O=picard_SRR27182345.bam  
RGLB=lib1 RGPL=illumina RGPU=run RGSM=SRR27182345 SORT_ORDER=coordinate CREATE_INDEX=true VALIDATION_STRINGENCY=LENIENT  
JavOpt:  
PicardProg: AddOrReplaceReadGroups
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ samtools faidx chr7.fa
```

```
krish@krISH17:/mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ ls
```

SRR27182345.sam	SRR27182345_2_fastqc.html	chr7.fa	chr7.fa.pac	picard_SRR27182345.bam	trim_SRR27182345_1_fastqc.html
SRR27182345_1.fastq	SRR27182345_2_fastqc.zip	chr7.fa.amb	chr7.fa.sa	pipeline.sh	trim_SRR27182345_1_fastqc.zip
SRR27182345_1_fastqc.html	adapter.fasta.txt	chr7.fa.ann	fastp.html	rmdup_SRR27182345.sam	trim_SRR27182345_2.fastq
SRR27182345_1_fastqc.zip	bwa_SRR27182345.sam	chr7.fa.bwt	fastp.json	sorted_SRR27182345.sam	trim_SRR27182345_2_fastqc.html
SRR27182345_2.fastq	chr7.dict	chr7.fa.fai	picard_SRR27182345.bai	trim_SRR27182345_1.fastq	trim_SRR27182345_2_fastqc.zip

```
krish@krISH17: /mnt/c/Users/KRISH SITIGAR/Desktop/EGFR_NGS$ java -jar ./gatk-4.3.0.0/gatk-package-4.3.0.0-local.jar HaplotypeCaller -R chr7.fa -I picard_SRR2
7182345.bam -O GATK_output.vcf
11:30:39.300 INFO NativeLibraryLoader - Loading libgkl_compression.so from jar:file:/mnt/c/Users/KRISH%20SITIGAR/Desktop/EGFR_NGS/gatk-4.3.0.0/gatk-package
-4.3.0.0-local.jar!/com/intel/gkl/native/libgkl_compression.so
11:30:39.630 INFO HaplotypeCaller - -----
11:30:39.631 INFO HaplotypeCaller - The Genome Analysis Toolkit (GATK) v4.3.0.0
11:30:39.631 INFO HaplotypeCaller - For support and documentation go to https://software.broadinstitute.org/gatk/
11:30:39.632 INFO HaplotypeCaller - Executing as krish@krISH17 on Linux v6.18.33.2-microsoft-standard-WSL2 amd64
11:30:39.632 INFO HaplotypeCaller - Java runtime: OpenJDK 64-Bit Server VM v21.0.11+10-1-24.04.2-Ubuntu
11:30:39.632 INFO HaplotypeCaller - Start Date/Time: July 29, 2026, 11:30:39 AM UTC
11:30:39.632 INFO HaplotypeCaller - -----
11:30:39.633 INFO HaplotypeCaller - -----
11:30:39.634 INFO HaplotypeCaller - HTSJDK Version: 3.0.1
11:30:39.634 INFO HaplotypeCaller - Picard Version: 2.27.5
11:30:39.634 INFO HaplotypeCaller - Built for Spark Version: 2.4.5
11:30:39.635 INFO HaplotypeCaller - HTSJDK Defaults.COMPRESSION_LEVEL : 2
11:30:39.635 INFO HaplotypeCaller - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
11:30:39.635 INFO HaplotypeCaller - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
11:30:39.635 INFO HaplotypeCaller - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
11:30:39.640 INFO HaplotypeCaller - Deflater: IntelDeflater
11:30:39.640 INFO HaplotypeCaller - Inflater: IntelInflater
11:30:39.640 INFO HaplotypeCaller - GCS max retries/reopens: 20
11:30:39.640 INFO HaplotypeCaller - Requester pays: disabled
11:30:39.640 INFO HaplotypeCaller - Initializing engine
11:30:39.927 INFO HaplotypeCaller - Done initializing engine
11:30:39.951 INFO HaplotypeCallerEngine - Disabling physical phasing, which is supported only for reference-model confidence output
11:30:39.966 INFO NativeLibraryLoader - Loading libgkl_utils.so from jar:file:/mnt/c/Users/KRISH%20SITIGAR/Desktop/EGFR_NGS/gatk-4.3.0.0/gatk-package-4.3.0
.0-local.jar!/com/intel/gkl/native/libgkl_utils.so
```